Updated TITE-CRM Likelihoods and Trial Design

Fixed-r CRM, random-r CRM, and the AIDE/IPDE recycle strategy

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7 Comparison of the updated likelihoods

For administration row i, let d_i be the assigned dose, z_i the IPDE indicator, and $w_i \in [0, 1]$ the TITE follow-up weight. Let

$$p_j(\alpha) = q_j^{\exp(\alpha)}$$

denote the power-CRM toxicity probability at dose j. The row-specific pre-TITE toxicity probability is denoted by θ_i. In both retained backends, the TITE-CRM contribution is

$$L_i = (w_i\theta_i)^{y_i}(1 - w_i\theta_i)^{1-y_i}.$$

Observed DLTs are assigned $w_i = 1$.

Backend	Pre-TITE model probability θ_i	Pending non-DLT term	Observed DLT term
Fixed-r CRM	$\theta_i = (1 - z_i)p_{d_i}(\alpha) + z_i\{r + (1 - r)p_{d_i}(\alpha)\}$, where r is fixed.	$1 - w_i\theta_i$	θ_i because $w_i = 1$
Random-r CRM	The same θ_i as fixed-r CRM, with $r \sim \mathrm{Beta}(a_r, b_r)$.	$1 - w_i\theta_i$	θ_i because $w_i = 1$

Common convention. A pending non-DLT contributes $1 - w_i\theta_i$. The delayed-outcome expression $(1 - \theta_i)^{w_i}$ is not used.

8 Trial design and recycle strategy

The simulation wrapper uses an event-clock design. Calendar time is tracked explicitly, and each treatment-administration row records its start time, evaluation time, dose, cycle number, cohort number, and regular-versus-retreat status.

Assignable events and waiting queue

An assignable event is an event that makes a treatment administration available for cohort filling:

1. A new patient arrival is an assignable event.
2. An IPDE/retreat decision at the end of a patient's evaluation window is also an assignable event. If the patient completes the previous evaluation without DLT and remains eligible for another cycle, a retreat event is scheduled at that evaluation time.

Both new-patient events and IPDE-available events are handled through the waiting queue. If an assignable event occurs while no cohort slot is open, the corresponding patient is added to the queue. The queue therefore contains:

- **new**: a newly arrived patient who has not yet received treatment;
- **retreat**: a previously treated patient who has completed evaluation and may be available for IPDE/recycling.

When cohort slots become available, newly arrived patients are selected before eligible retreat/IPDE patients. Within each patient type, events are ordered by event time, with a sequence number used to break ties.

Suspension rule

Let n_{eval} denote the minimum number of evaluated patients required before model-based assignment can proceed. If the number of evaluated patients is smaller than n_{eval}, the trial is suspended. During suspension, no treatment assignment or cohort formation is performed. All newly arriving patients and all patients who become eligible for IPDE/retreat are added directly to the waiting queue. The trial resumes once the number of evaluated patients reaches n_{eval}, after which the model-based dose decision and the usual cohort-filling procedure are applied.

Cohort formation and model-based decisions

A cohort remains open until `cohortsize` administration rows have been assigned. Provided that the suspension rule is not active, the algorithm repeatedly fills available slots from the waiting queue, giving priority to new arrivals and then using eligible retreat/IPDE patients for any remaining slots.

Once the cohort is full, it is closed. The design waits until another assignable event occurs. At that time, the interim dataset is reconstructed using all administrations started by the decision time, their observed DLT indicators, and their TITE weights. The selected CRM backend is fitted with the common TITE-CRM likelihood, and the next dose is selected subject to the no-skipping and dose-cap rules.

After the model returns the next dose, a new cohort is opened and the waiting queue is used again, provided that the suspension rule is not active. The trial is therefore event-driven: decisions are triggered by assignable events, the evaluated-sample requirement, and cohort availability rather than by fixed calendar intervals.

Recycle eligibility and cohort priority

A retreat/IPDE event can be assigned only when the following conditions and priority rule are satisfied:

1. The patient has a previous administration record.
2. The most recent administration was non-DLT.
3. The most recent administration has completed the DLT assessment window.
4. The patient's number of cycles is below `cycle_max`.
5. The current dose determined by the model is above dose 1.
6. When both new arrivals and eligible retreat/IPDE patients are waiting during cohort formation, new arrivals are assigned first.

The previous requirement that the patient's most recent dose equal the immediately lower dose, $d_{\text{last}} = d_{\text{current}} - 1$, is not imposed. Thus, an otherwise eligible patient may be recycled when the current dose determined by the model is above dose 1, regardless of whether the most recent dose was exactly one level lower.

Eligible retreat/IPDE patients are used only after no new arrivals remain available for the open cohort. Retreat events that are not currently eligible remain in the queue until they become eligible or the trial ends.